



NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPPALLI – 620 015
DEPARTMENT OF CIVIL ENGINEERING
B.Tech. / I Semester (2025 July-Nov session)
CEIR11 / Basics of Civil Engineering/ I year ICE-B

Cycle Test-2

Date: 27 Nov 2025 (Thursday) Time: 04:30 PM to 5:30 PM Max Marks: 20

Name: [REDACTED] Roll No: [REDACTED]

Instructions:

- There are totally 9 questions. Read all the questions carefully before answering.
- Draw/Write your answers clearly and legibly. Make necessary assumptions if not given.
- Support your answers with suitable figures wherever necessary.

S.No	Questions	Marks	CO's
Q.1:	What are the principles of surveying. ✓	2	CO4
Q.2:	List all the equipments (with purpose) used in Plane table surveying. ✓	2	CO4
Q.3:	What is whole circle bearing and reduced bearings. ✓	2	CO4
Q.4:	List the process involved in self purification of streams. ✓	2	CO5
Q.5:	What is the desirable and permissible limits of fluoride in water and suggest a treatment method if it exceeds the limit. ✓x	2	CO5
Q.6:	What are the physical characteristic of a drinking water. Explain. x ✓	2	CO5
Q.7:	Explain radiation method using a compass to calculate the area of circular plot.	2	CO4
Q.8:	Explain the procedure to calculate the area (Figure.1) of a building using offset method using Chain. ✓	3	CO4
Q.9:	Draw and label the flow diagram of water treatment. ✓	3	CO5



End of the Question Paper

